

Supplementary Material

For “Learning Selective Invariance for Augmentation-based Self-Supervised Learning to Improve Cross-User Human Activity Recognition”

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I. AUGMENTATION METHODS

Inspired by [1], we adapt two adversarial amplitude-based augmentations originally developed for images (AAUA and AAD) to time series; details are provided below.

A. AAD: Gradient-guided Dropout on Non-critical Frequencies

AAD uses gradients with respect to the amplitude as a surrogate sensitivity map and drops highly relied components. We compute amplitude gradients

$$g_A(f, c) = \nabla_{A(f, c)} \mathcal{L}. \quad (S1)$$

We apply min-max normalization $s(\cdot)$ to obtain $\bar{g}_A = s(g_A)$ and sample a random threshold $p(f, c) \sim \mathcal{U}(0, 1)$. The dropout mask is defined as

$$M_{\text{AAD}}(f, c) = \mathbb{I}(\bar{g}_A(f, c) < p(f, c)), \quad (S2)$$

and the dropped amplitude is

$$A^{\text{AAD}}(f, c) = A(f, c) \odot M_{\text{AAD}}(f, c). \quad (S3)$$

The augmented view is reconstructed with the original phase Φ via $\tilde{x} = \mathcal{F}^{-1}(A^{\text{AAD}}, \Phi)$.

B. AAUA: Uncertainty-aware Adversarial Update on Critical-unstable Frequencies

AAUA perturbs the amplitude statistics in an uncertainty-aware manner. Let $\mu(A) \in \mathbb{R}^C$ and $\sigma(A) \in \mathbb{R}^C$ denote the channel-wise mean and standard deviation of amplitudes. AAUA estimates the uncertainty of these statistics by their batch-wise standard deviations $\Sigma_\mu, \Sigma_\sigma \in \mathbb{R}^C$ and updates the augmentation statistics as

$$\mu'(A) = \mu(A) + d_t^\mu \odot \Sigma_\mu, \quad \sigma'(A) = \sigma(A) + d_t^\sigma \odot \Sigma_\sigma, \quad (S4)$$

where d_t^μ, d_t^σ are adversarial control variables. Using instance-normalization style reparameterization, AAUA reconstructs an augmented amplitude on the selected region as

$$A^{\text{AAUA}} = \left(\frac{A - \mu(A)}{\sigma(A)} \right) \odot \sigma'(A) + \mu'(A), \quad (S5)$$

while keeping the complementary region unchanged, consistent with the original formulation. The adversarial variables are updated by a sign gradient step

$$d_{t+1}^\mu = d_t^\mu + \text{sign}(\nabla_{d_t^\mu} \mathcal{L}), \quad d_{t+1}^\sigma = d_t^\sigma + \text{sign}(\nabla_{d_t^\sigma} \mathcal{L}), \quad (S6)$$

and the resulting view is reconstructed with the original phase.

II. ARCHITECTURE-LEVEL ANALYSIS

To further clarify how the proposed spectral gate interacts with different self-supervised architectures, we conduct additional experiments on SHAR under the same cross-user transfer protocol and linear evaluation setting as the main paper. Following the main experiments, we use FreRA [2] as the SimCLR-based baseline, and denote our method under this framework as SimCLR+Gate. We further plug the same spectral gating module into two additional SSL frameworks, BYOL and SimSiam, to evaluate whether the proposed mechanism depends on a specific contrastive objective or prediction head.

As shown in Table S1, SimCLR+Gate improves over the SimCLR/FreRA baseline from 52.5% to 57.7% on average. The same gate also improves BYOL from 52.2% to 55.1% and achieves comparable performance under SimSiam. These results suggest that the proposed spectral gating mechanism is not restricted to a single SSL architecture, but can serve as a plug-in augmentation module for different contrastive and non-contrastive self-supervised frameworks.

REFERENCES

- [1] X. He, J. Hu, Q. Lin, C. Luo, W. Xie, S. Song, M. H. Khan, and L. Shen, “Towards combating frequency simplicity-biased learning for domain generalization,” *Advances in Neural Information Processing Systems*, vol. 37, pp. 31 078–31 102, 2024.
- [2] T. Tian, C. Miao, and H. Qian, “Frera: a frequency-refined augmentation for contrastive learning on time series classification,” in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*, 2025, pp. 2835–2846.

TABLE S1
ARCHITECTURE-LEVEL ANALYSIS ON SHAR UNDER THE SD=3 TRANSFER SETTING. RESULTS ARE TARGET-DOMAIN CLASSIFICATION ACCURACY (%).

SD	TD	Methods					
		SimCLR	SimCLR+Gate	BYOL	BYOL+Gate	SimSiam	SimSiam+Gate
3	1	60.2	60.7	53.4	55.2	54.4	54.9
	2	46.7	62.3	52.8	56.4	49.4	53.0
	3	66.5	65.8	61.5	66.8	58.9	54.6
	5	36.6	41.9	40.9	41.9	39.9	41.3
	avg	52.5	57.7	52.2	55.1	50.7	51.0